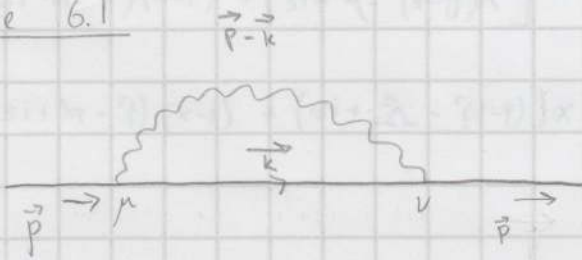


Exercise 6.1



The external legs of this diagram are part of a bigger diagram so we label them as propagators which gives the Feynman rule:

Dirac propagator: $\frac{i(\not{p}+m)}{p^2-m^2+i\epsilon}$

In the diagram we have loop of photon which means we need to integrate over all the possible momentum k , we get the k -integral integrating the whole thing. The vertex is standard QED-vertex that gives us $-ie\gamma^\mu$. Using the Feynman rules we get

$$= \int \frac{d^4k}{(2\pi)^4} \frac{i(\not{p}+m)}{p^2-m^2+i\epsilon} (-ie\gamma^\mu) \frac{i(\not{k}+m)}{k^2-m^2+i\epsilon} (-ie\gamma^\nu) \frac{i(\not{p-k}+m)}{(p-k)^2+i\epsilon} \frac{-ig^{\mu\nu}}{(p-k)^2+i\epsilon}$$

We don't need $i\epsilon$ prescription for fermion propagator whose momenta we aren't integrating over so we can drop them

Photon propagator

The expression for the diagrams is

$$= \frac{i(\not{p}+m)}{p^2-m^2} \left[-e^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\mu \frac{(\not{k}+m)}{k^2-m^2+i\epsilon} \gamma^\nu \frac{1}{(p-k)^2+i\epsilon} \right] \frac{i(\not{p-k}+m)}{(p-k)^2+i\epsilon}$$

Exercise 6.2

$$-i\Sigma_2(p) \equiv -e^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\mu \frac{(\not{k}+m)}{k^2-m^2+i\epsilon} \gamma^\nu \frac{1}{(p-k)^2+i\epsilon}$$

Let's regulate IR divergence by small photon mass μ and UV-divergence by Pauli-Villars:

$$\frac{1}{(p-k)^2+i\epsilon} \rightarrow \frac{1}{(p-k)^2-\mu^2+i\epsilon} - \frac{1}{(p-k)^2-\Lambda^2+i\epsilon}$$

$$\Rightarrow -i\Sigma_2(p) = -e^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\mu (\not{k}+m) \gamma^\nu \frac{1}{k^2-m^2+i\epsilon} \left[\frac{1}{(p-k)^2-\mu^2+i\epsilon} - \frac{1}{(p-k)^2-\Lambda^2+i\epsilon} \right]$$

Now let's use Feynman parametrization: $\frac{1}{AB} = \int_0^1 dx \frac{1}{[xA + (1-x)B]^2}$

$$\Rightarrow -i \Sigma_2(p) = -e^2 \int \frac{d^4 k}{(2\pi)^4} \int_0^1 dx \gamma_\mu (k+m) \gamma^\mu \left[\frac{1}{x((p-k)^2 - \mu^2 + i\varepsilon) + (1-x)(k^2 - m^2 + i\varepsilon)} \right]^{-2} \\ - \left[\frac{1}{x((p-k)^2 - \mu^2 + i\varepsilon) + (1-x)(k^2 - m^2 + i\varepsilon)} \right]^{-2}$$

Let's simplify some of the expressions:

$$\gamma_\mu (k+m) \gamma^\mu = -2k + 4m$$

$$x[(p-k)^2 - \mu^2 + i\varepsilon] + (1-x)(k^2 - m^2 + i\varepsilon) = x p^2 + x k^2 - 2x p \cdot k - \mu^2 x + x i\varepsilon + k^2 - x k^2 - (1-x)m^2 + i\varepsilon - x i\varepsilon \\ = i\varepsilon + k^2 - 2x p \cdot k + x^2 p^2 - x^2 p^2 + x p^2 - \mu^2 x - m^2(1-x) = (k - x p)^2 + x(1-x)p^2 - (1-x)m^2 - \mu^2 x + i\varepsilon$$

$$\text{let's define } (k - x p) = l \quad \text{and} \quad \Delta \equiv -x(1-x)p^2 + (1-x)m^2 + \mu^2 x$$

$$\Rightarrow x[(p-k)^2 - \mu^2 + i\varepsilon] + (1-x)(k^2 - m^2 + i\varepsilon) = l^2 - \Delta + i\varepsilon$$

$$\text{the shift of } l = k - x p \text{ means } dk = dl$$

$$\text{Now we simplify: } x[(p-k)^2 - \mu^2 + i\varepsilon] + (1-x)(k^2 - m^2 + i\varepsilon) = x p^2 + x k^2 - 2x p \cdot k - \mu^2 x + x i\varepsilon + k^2 - x k^2 - m^2(1-x) + i\varepsilon - x i\varepsilon \\ = i\varepsilon + k^2 - 2x p \cdot k + x^2 p^2 - x^2 p^2 + x p^2 - \mu^2 x - m^2(1-x) = l^2 - \Delta + i\varepsilon$$

$$\text{where } \Delta = -x(1-x)p^2 + (1-x)m^2 + \mu^2 x \quad (\text{also } k = l + x p)$$

$$\Rightarrow -i \Sigma_2(p) = -e^2 \int \frac{d^4 l}{(2\pi)^4} \int_0^1 dx (-2(l + x p) + 4m) \left[\frac{1}{(l^2 - \Delta + i\varepsilon)^2} - \frac{1}{(l^2 - \Delta + i\varepsilon)^2} \right]$$

The term in numerator proportional to l integrates to zero so we can set it zero.

$$\Rightarrow -i \Sigma_2(p) = -e^2 \int_0^1 dx \int \frac{d^4 l}{(2\pi)^4} \frac{-2x p + 4m}{(l^2 - \Delta + i\varepsilon)^2} - \frac{-2x p + 4m}{(l^2 - \Delta + i\varepsilon)^2}$$

$$\text{Now we Wick rotate via } l^0 \rightarrow i l_E^0 \text{ and } l^i = l_E^i \Rightarrow dl^0 \rightarrow i dl_E^0$$

$$\Rightarrow \frac{d^4 l}{(2\pi)^4} \xrightarrow{\text{Minkowski}} i \frac{d^4 l_E}{(2\pi)^4} \xrightarrow{\text{Euclid}} \quad ||l^2 = l^0^2 - l^i^2 \rightarrow -l_E^2$$

$$\Rightarrow -i \Sigma_2(p) = -e^2 \int_0^1 dx i \int \frac{d^4 l_E}{(2\pi)^4} \frac{-2x p + 4m}{(-l_E^2 - \Delta + i\varepsilon)^2} - \frac{-2x p + 4m}{(-l_E^2 - \Delta + i\varepsilon)^2}$$

To evaluate the Euclidean integral we no-longer need $i\varepsilon$ -prescription so we can set it to 0

$$\Rightarrow \Sigma_2(p) = 2e^2 \int_0^1 dx (-x p + 2m) \int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{(l_E^2 + \Delta)^2} - \frac{1}{(l_E^2 + \Delta)^2}$$

$$\text{We can now use the spherical coordinates } \int d^4 l_E = \int d\Omega_4 \int_0^\infty dl_E l_E^3$$

$$\text{with } \int d\Omega_4 = 2\pi^2 \quad \text{and} \quad \int_0^\infty dl_E l_E^3 \frac{1}{(l_E^2 + \Delta)^2} = \frac{1}{2} \left(\frac{\Delta}{\Delta + l_E^2} + \ln(l_E^2 + \Delta) \right) \Big|_{l_E=0}^{l_E=\infty}$$

For the term $\frac{1}{(l_E^2 + \Delta)^2}$ we get same answer but with Δ_A instead of Δ using the result directly

$$\Rightarrow \Sigma_2(p) = \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx (-x\not{p} + 2m) \frac{1}{2} \left[\frac{\Delta}{\Delta + l_E^2} \Big|_{l_E=0}^{l_E=\infty} - \frac{\Delta \Delta}{\Delta \Delta + l_E^2} \Big|_{l_E=0}^{l_E=\infty} + \ln \left[\frac{l_E^2 + \Delta}{l_E^2 + \Delta \Delta} \right] \Big|_{l_E=0}^{l_E=\infty} \right]$$

$$\lim_{l_E \rightarrow \infty} \ln \left(\frac{l_E^2 + \Delta}{l_E^2 + \Delta \Delta} \right) = \ln(1) = 0, \quad \lim_{l_E \rightarrow \infty} \frac{\Delta}{\Delta + l_E^2} = \frac{\Delta}{\infty} = 0$$

$$\Rightarrow \Sigma_2(p) = \frac{1}{8\pi^2} e^2 \int_0^1 dx (-x\not{p} + 2m) \left(-\ln \left[\frac{\Delta}{\Delta \Delta} \right] \right) \quad \parallel \quad \frac{e^2}{4\pi} = \alpha$$

$$\Sigma_2(p) = \frac{\alpha}{2\pi} \int_0^1 dx (-x\not{p} + 2m) \ln \left[\frac{\Delta \Delta}{\Delta} \right]$$

$\Delta \Delta = x\Delta^2 + (1-x)m^2 - x(1-x)p^2$ So in limit of $\Delta \rightarrow \infty$ the leading contribution is $x\Delta^2$. We can keep only this contribution when looking at UV divergence since $\Delta \rightarrow \infty$ as a result we get:

$$\Sigma_2(p) = \frac{\alpha}{2\pi} \int_0^1 dx (-x\not{p} + 2m) \ln \left[\frac{x\Delta^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2} \right] \quad \square$$

Exercise 6.3



The first diagram is $\frac{i\not{p} + m}{p^2 - m^2}$

Second diagram is $\frac{i\not{p} + m}{p^2 - m^2} [-i\Sigma_2(p)] \frac{i\not{p} + m}{p^2 - m^2}$

We can factor the third diagram as a "multiplication" of two second diagrams

$$\frac{i\not{p} + m}{p^2 - m^2} [-i\Sigma_2(p)] \frac{i\not{p} + m}{p^2 - m^2} [-i\Sigma_2(p)] \frac{i\not{p} + m}{p^2 - m^2}$$

We can use $i\not{p} \Sigma_2(p) = i\Sigma_2(p) \not{p}$ since $\not{p}^2 = p^2$ and only Gamma-matrix term in $\Sigma_2(p)$ is $\not{p}$ so $[\Sigma_2(p), \not{p}] = 0$

$$\Rightarrow \text{Third diagram} = \left[\frac{i\not{p} + m}{p^2 - m^2} [-i\Sigma_2(p)] \frac{i\not{p} + m}{p^2 - m^2} \right]^2$$

And if we have fourth diagram then we just get one more $i\Sigma_2(p) \frac{i\not{p} + m}{p^2 - m^2}$ term times previous 3-loop diagram so this progresses as geometric series.

A geometric series can be written as $a + aq + aq^2 + aq^3 + aq^4 + \dots = a \sum_{n=0}^{\infty} q^n$
 $\Rightarrow a \sum_{n=0}^{\infty} q^n = \frac{a}{1-q}$ for $|q| < 1$, we assume $|q| < 1$ since we are describing

a physical process.

$$\Rightarrow \text{diagram} + \text{diagram} + \text{diagram} + \dots = \frac{i(\not{p}+m)}{p^2-m^2} \left[1 - (-i \cdot i \Sigma_2(p) \frac{(\not{p}+m)}{p^2-m^2}) \right]^{-1}$$

$$\frac{1}{\not{p}-m} = \frac{(\not{p}+m)}{p^2-m^2} \Rightarrow \text{diagram} + \text{diagram} + \dots = \frac{i}{\not{p}-m} \cdot \frac{1}{1 - \frac{\Sigma_2(p)}{\not{p}-m}}$$

$$\Rightarrow = \frac{i}{\not{p}-m - \Sigma_2(p)}$$

From given definitions for $\Sigma'(p^2)$ and $\Sigma''(p^2)$ and the form we calculated for

$\Sigma_2(p)$ we see that $\Sigma_2(p) = \not{p} \Sigma'(p^2) + m \Sigma''(p^2)$

$$\Rightarrow \frac{i}{\not{p}-m-\Sigma_2(p)} = \frac{i}{\not{p}-m-\not{p}\Sigma'(p^2)-m\Sigma''(p^2)} = \frac{i}{\not{p}(1-\Sigma'(p^2))-m(1+\Sigma''(p^2))}$$

$$= \frac{i}{\not{p}^2(1-\Sigma'(p^2))^2 - m^2(1+\Sigma''(p^2))^2} \quad \parallel \quad \not{p}^2 = p^2$$

$$\frac{i}{\not{p}-m-\Sigma_2(p)} = i \frac{\not{p}(1-\Sigma'(p^2)) + m(1+\Sigma''(p^2))}{p^2(1-\Sigma'(p^2))^2 - m^2(1+\Sigma''(p^2))^2} \quad \parallel \quad \text{No idea how the } i \text{ is supposed to disappear.}$$

Exercise 6.4

The physical mass $p^2 = m_e^2$ defined as the pole of the Propagator:

The pole is: $p^2(1-\Sigma'(p^2))^2 \Big|_{p^2=m_e^2} - m^2(1+\Sigma''(p^2))^2 = 0$

$\Sigma' \propto \alpha$ and $\Sigma'' \propto \alpha$ so we can look at this to the leading order in α
 since α is small: $\hookrightarrow$ Taylor: $(1+x)^\alpha \approx 1+\alpha x$ for small x

$$m_e^2 = m^2 (1 + \Sigma''(p^2))^2 (1 - \Sigma'(p^2))^2 \approx m^2 [1 + 2\Sigma''(p^2) + \mathcal{O}(\alpha^2)] [1 + 2\Sigma'(p^2) + \mathcal{O}(\alpha^2)]$$

$$= m^2 [1 + 2\Sigma''(p^2) + 2\Sigma'(p^2) + \mathcal{O}(\alpha^2)] \quad \parallel \quad \begin{aligned} \text{Now insert } \Sigma' &= -\frac{\alpha}{2\pi} \int_0^1 dx \ln(\dots) \\ \Sigma'' &= \frac{\alpha}{\pi} \int_0^1 dx \ln(\dots) \end{aligned}$$

$$m_e^2 = m^2 \left[1 + \frac{\alpha}{2\pi} \int_0^1 dx [4-2x] \ln \left[\frac{x-x^2}{-x(1-x)p^2 + (1-x)m^2 + x p^2} \right] \right]$$

To the leading order in α we can write $p^2 \approx m^2$ inside the integral
 $\Rightarrow -x(1-x)p^2 + (1-x)m^2 \approx (1-x)^2 m^2$

$$\Rightarrow m_e^2 = m^2 \left[1 + \frac{\alpha}{2\pi} \int_0^1 dx [4-2x] \ln \left(\frac{x\Lambda^2}{(1-x)^2 m^2 + x\mu^2} \right) \right]$$

Near the pole the propagator has the form

$$Z_2 \frac{i(\not{p}+m)}{p^2-m^2}$$

Since Σ and δ is small we get:

$$\frac{i}{\not{p}-m-\Sigma_2(p)} \approx \frac{i}{(\not{p}-m)(1-\frac{\partial \Sigma_2}{\partial \not{p}}|_{m_e})} = \frac{i(\not{p}+m)}{p^2-m^2} \frac{1}{1-\frac{\partial \Sigma_2(p)}{\partial \not{p}}|_{m_e}}$$

$$\Rightarrow Z_2 = \left(1 - \frac{\partial \Sigma_2(p)}{\partial \not{p}}|_{m_e}\right)^{-1} \approx 1 + \frac{\partial \Sigma_2(p)}{\partial \not{p}}|_{m_e} = 1 + \frac{\partial \Sigma_2(p)}{\partial \not{p}}|_m \text{ since}$$

We can approximate m inside the dx integral to the first order in α .

$$\frac{\partial \Sigma_2(p)}{\partial \not{p}} = \frac{\partial}{\partial \not{p}} \left[\frac{\alpha}{2\pi} \int_0^1 dx (-x\not{p}+2m) \ln \left[\frac{x\Lambda^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2} \right] \right]$$

$$p^2 = \not{p}\not{p} = p^2 \quad \text{and} \quad \frac{d}{dx} \ln \left[\frac{a}{bx+j} \right] = \frac{d}{dx} [\ln a - \ln [bx+j]] = -\frac{\frac{d}{dx}(bx)}{bx+j}$$

$$\Rightarrow \frac{\partial \Sigma_2(p)}{\partial \not{p}}|_m = \frac{\alpha}{2\pi} \left[- \int_0^1 dx x \ln \left[\frac{x\Lambda^2}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2} \right] - (-x\not{p}+2m) \frac{2(-x(1-x))\not{p}}{-x(1-x)p^2 + (1-x)m^2 + x\mu^2} \right] \Big|_m$$

$$\text{Boundary condition } \not{p}=m \Rightarrow (-x\not{p}+2m)\not{p}|_m = (2-x)m^2$$

$$\Rightarrow Z = 1 + \frac{\alpha}{2\pi} \int_0^1 dx \left[-x \ln \left(\frac{x\Lambda^2}{(1-x)^2 m^2 + x\mu^2} \right) + (2-x) \frac{2m^2 x(1-x)}{(1-x)^2 m^2 + x\mu^2} \right]$$

$$\Rightarrow Z = 1 + \frac{\alpha}{2\pi} \int_0^1 dx \left[-x \ln \left(\frac{x\Lambda^2}{(1-x)^2 m^2 + x\mu^2} \right) + (2-x) \frac{2m^2 x(1-x)}{(1-x)^2 m^2 + x\mu^2} \right]$$

The first integral in expression of Z is UV-divergent so we can approximate the logarithm as $\ln \left(\frac{\Lambda^2}{m^2} \right)$, which gives us

$$\frac{\alpha}{2\pi} \int_0^1 dx (-x \ln \left(\frac{\Lambda^2}{m^2} \right)) = -\frac{1}{2} \frac{\alpha}{2\pi} (1^2 - 0^2) \ln \frac{\Lambda^2}{m^2} = -\frac{1}{2} \frac{\alpha}{2\pi} \ln \left[\frac{\Lambda^2}{m^2} \right]$$

$$\text{The second integral is } IR = \frac{\alpha}{2\pi} 2m^2 \int_0^1 dx (1+(1-x)) \frac{(1-(1-x))(1-x)}{(1-x)^2 m^2 + x\mu^2}$$

$$IR = \frac{\alpha}{\pi} m^2 \int_0^1 dx \frac{[1-(1-x)^2](1-x)}{(1-x)m^2 + x\mu^2} = \frac{\alpha}{\pi} \int_0^1 dx \frac{1-x}{(1-x)^2 + x \frac{\mu^2}{m^2}} [1 + \mathcal{O}(1-x)]$$

Let's evaluate this to the first order in $(1-x)$, since $1-x$ is small as $x \rightarrow 1$. and at $x=0$ our integral is just zero because integrand is $(2-x) \frac{2m^2 x(1-x)}{(1-x)^2 m^2 + x\mu^2}$

$$\Rightarrow IR \approx \frac{\alpha}{\pi} \int_0^1 dx \frac{1-x}{(1-x)^2 + x \frac{\mu^2}{m^2}}$$

$$(1-x)^2 + x \frac{\mu^2}{m^2} = 1 - 2x + x^2 + x \frac{\mu^2}{m^2} = x^2 + x \left(\frac{\mu^2}{m^2} - 2 \right) + 1$$

This has root at

$$x_0 = \frac{-\left(\frac{\mu^2}{m^2} - 2\right) \pm \sqrt{\left(\frac{\mu^2}{m^2} - 2\right)^2 - 4}}{2}$$

$$x_{01} = -\left(\frac{\mu^2}{2m^2} - 1\right) - \frac{1}{2} \sqrt{\left(\frac{\mu^2}{m^2} - 2\right)^2 - 4}, \quad x_{02} = -\left(\frac{\mu^2}{2m^2} - 1\right) + \frac{1}{2} \sqrt{\left(\frac{\mu^2}{m^2} - 2\right)^2 - 4}$$

$$\Rightarrow (1-x)^2 + x \frac{\mu^2}{m^2} = \left(x + \left[\frac{\mu^2}{2m^2} - 1\right] - \frac{1}{2} \sqrt{\left(\frac{\mu^2}{m^2} - 2\right)^2 - 4}\right) \left(x + \left[\frac{\mu^2}{2m^2} - 1\right] + \frac{1}{2} \sqrt{\left(\frac{\mu^2}{m^2} - 2\right)^2 - 4}\right)$$

Let's do change of variables: $\left(\frac{\mu^2}{m^2} - 2\right) \equiv b$ and $x + \frac{1}{2}b = u \Rightarrow x = u - \frac{1}{2}b$

$\Rightarrow du = dx$ and u has ^{integration} limits from $0 + \frac{1}{2}b$ to $1 + \frac{1}{2}b$

$$\Rightarrow IR = \frac{\alpha}{\pi} \int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \frac{1 - u + \frac{1}{2}b}{\left(u - \frac{1}{2}\sqrt{b^2 - 4}\right) \left(u + \frac{1}{2}\sqrt{b^2 - 4}\right)} \quad \parallel \quad k \equiv \frac{1}{2}\sqrt{b^2 - 4}$$

$$= -\frac{\alpha}{\pi} \int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \frac{1 - u + \frac{1}{2}b}{(k-u)(k+u)} \quad \parallel \quad \frac{1}{(k-u)(k+u)} = +\frac{1}{2k} \left[\frac{1}{k-u} + \frac{1}{k+u} \right]$$

$$= -\frac{\alpha}{2\pi k} \int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du \frac{(1 + \frac{1}{2}b - u)}{(k+u)} \left[\frac{1}{k+u} + \frac{1}{k-u} \right]$$

$$= -\frac{\alpha}{2\pi k} \left(1 + \frac{1}{2}b\right) \left[\ln[k+u] - \ln[k-u] \right] \Big|_{\frac{1}{2}b}^{1+\frac{1}{2}b} + \frac{\alpha}{2\pi k} \int_{\frac{1}{2}b}^{1+\frac{1}{2}b} du u \left[\frac{1}{k+u} + \frac{1}{k-u} \right]$$

$$\ln(k+u) = \ln\left(k\left(1 + \frac{u}{k}\right)\right) = \ln(k) + \ln\left(1 + \frac{u}{k}\right) \quad \text{and}$$

$$\ln(k-u) = \ln\left(k\left(1 - \frac{u}{k}\right)\right) = \ln(k) + \ln\left(1 - \frac{u}{k}\right)$$

$$\Rightarrow \ln(k+u) - \ln(k-u) = \ln\left(1 + \frac{u}{k}\right) - \ln\left(1 - \frac{u}{k}\right)$$

$$\text{and} \quad \int du \frac{u}{k+u} + \frac{u}{k-u} = -k \ln(k+u) + u - k \ln(k-u) - u = -k \ln(k^2 - u^2)$$

$$\Rightarrow IR = -\frac{\alpha}{2\pi} \left(1 + \frac{1}{2}b\right) \left[\ln\left(1 + \frac{1+\frac{1}{2}b}{k}\right) - \ln\left(1 + \frac{\frac{1}{2}b}{k}\right) - \ln\left[1 - \frac{1+\frac{1}{2}b}{k}\right] + \ln\left[1 - \frac{\frac{1}{2}b}{k}\right] \right] \\ - \frac{\alpha}{2\pi} \left[\ln\left(\frac{k^2 - (1+\frac{1}{2}b)^2}{k^2 - (\frac{1}{2}b)^2}\right) \right]$$

First, let's investigate the term what happens at infrared (IR)-divergence:
 $\mu \rightarrow 0$, let's start by investigating term $\frac{1+\frac{1}{2}b}{k}$

$$\frac{1 + \frac{1}{2}b}{k} = \frac{1 + \frac{1}{2}b}{\frac{1}{2}\sqrt{b^2-4}} = \sqrt{\frac{(2+b)^2}{(b+2)(2-b)}} = \sqrt{\frac{2+b}{2-b}}$$

$$\text{IR-divergence } \mu \rightarrow 0 \Rightarrow b \rightarrow -2 \Rightarrow \lim_{b \rightarrow -2} \sqrt{\frac{2+b}{2-b}} = \sqrt{\frac{0}{4}} = 0$$

While this term does multiply 4-logarithms we must first see what happens to the logarithms as $b \rightarrow -2$ since if they diverge we need to investigate further:

$$\ln\left(1 + \frac{1 + \frac{1}{2}b}{k}\right) = \ln(1) = 0 \quad \text{similarity: } \ln\left(1 - \frac{1 + \frac{1}{2}b}{k}\right) = 0$$

$\frac{1 + \frac{1}{2}b}{k} \xrightarrow{b \rightarrow -2} 0$

$$\ln\left(1 + \frac{\frac{1}{2}b}{k}\right) = \ln\left(1 - k + \frac{1 + \frac{1}{2}b}{k}\right) = \ln(1-k) \quad \parallel \text{ and } k = \frac{1}{2}\sqrt{b^2-4} = 0 \text{ as } b \rightarrow 0$$

$$\ln\left(1 + \frac{\frac{1}{2}b}{k}\right) = \ln(1) = 0 \quad \text{and} \quad \ln\left(1 - \frac{\frac{1}{2}b}{k}\right) = \ln\left(1 + k - \frac{1 + \frac{1}{2}b}{k}\right) = \ln(1) = 0$$

so all the logarithms that multiply $\frac{1}{2\pi} \frac{1 + \frac{1}{2}b}{k}$ are 0 so there is no

bad behaviour on the first part of the expression IR
now let's investigate the logarithm

$$\begin{aligned} \ln\left[\frac{k^2 - (1 + \frac{1}{2}b)^2}{k^2 - (\frac{1}{2}b)^2}\right] &= \ln\left[\frac{k^2 - (1 + \frac{1}{2}b + \frac{1}{4}b^2)}{k^2 - \frac{b^2}{4}}\right] \quad \parallel k^2 = \frac{1}{4}(b^2 - 4) = \frac{b^2}{4} - 1 \\ &\Rightarrow = \ln\left[\frac{-2-b}{-1}\right] = \ln[b+2] \quad \parallel b = \frac{\mu^2}{m^2} - 2 \\ &= \ln\left[\frac{\mu^2}{m^2}\right] \end{aligned}$$

we have found the IR-divergence of the Z_2

$$\Rightarrow Z_2 = 1 - \frac{1}{2} \frac{\alpha}{2\pi} \ln\left[\frac{\Lambda^2}{m^2}\right] + \text{IR} = 1 - \frac{1}{2} \frac{\alpha}{2\pi} \ln\left[\frac{\Lambda^2}{m^2}\right] - \frac{\alpha}{2\pi} \ln\left[\frac{\mu^2}{m^2}\right]$$

$$Z_2 = 1 + \frac{\alpha}{2\pi} \left[-\frac{1}{2} \ln \frac{\Lambda^2}{m^2} + \ln \frac{m^2}{\mu^2} \right]$$

According to LSZ-theorem each external leg comes with $\sqrt{Z_2}$

$$\Rightarrow \text{Cross section get } Z_2^2 \approx 1 + \frac{\alpha}{2\pi} \left(-\ln \frac{\Lambda^2}{m^2} + 2 \ln \frac{m^2}{\mu^2} \right) + \text{Finite to } \mathcal{O}(\alpha^2)$$

(we ignore terms of order α^2)

$$\text{So the external leg cross section becomes } d\sigma^{\text{ext}} = d\sigma^0 Z_2^2 = d\sigma^0 \frac{\alpha}{2\pi} \left[-\ln \frac{\Lambda^2}{m^2} + 2 \ln \frac{m^2}{\mu^2} \right] + \text{finite terms}$$

$$\text{We also have } d\sigma^{\text{rad}} + d\sigma^{\text{vertex}} = \text{Finite} + d\sigma^0 \cdot \frac{\alpha}{2\pi} \left(\ln \left(\frac{\Lambda^2}{m^2} \right) - 2 \ln \left(\frac{m^2}{\mu^2} \right) \right) \quad (\text{from last week})$$

so we get

$$d\sigma^{\text{ext}} + d\sigma^{\text{rad}} + d\sigma^{\text{vertex}} = \text{finite} + d\sigma^0 \cdot \frac{\alpha}{2\pi} \left(-\ln \left(\frac{\Lambda^2}{m^2} \right) + 2 \ln \left(\frac{m^2}{\mu^2} \right) \right) + d\sigma^0 \frac{\alpha}{2\pi} \left(\ln \left(\frac{\Lambda^2}{m^2} \right) - 2 \ln \left(\frac{m^2}{\mu^2} \right) \right)$$

$\Rightarrow$ The infinite terms cancel each other.

$$\Rightarrow d\sigma^{\text{rad}} + d\sigma^{\text{virt}} + d\sigma^{\text{ext}} = \text{finite number} \quad \square$$

$\Rightarrow$ We have shown that the UV and IR divergence from the expression of Z_2 leads to finite expression for cross-section.